

Response to Reviewers

KinaseCanvas: An Integrated Resource for Curated and Novel Kinase-Substrate Interactions

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We thank both reviewers for their thorough, constructive reading of the manuscript. Their comments led us to re-audit the data pipeline in detail, turning up two implementation errors that had not previously been caught. These are described in full below, since one directly caused a case study Reviewer 2 correctly identified as implausible. The released resource has been regenerated, and every figure and statistic in the manuscript reflects the corrected pipeline. Reviewer comments are quoted, and where we have added or changed manuscript text in response we quote the exact new text and say where it now lives.

Reviewer 1

1.

Reliance on old-fashioned peptide-library screens for substrate prediction matching is a weakness. If one looks at the original images from these studies, one can see that signal-to-noise is quite marginal for many of the kinases profiled, not just ones billed as ‘promiscuous’. A much better approach is the one described by Sugiyama et al. (PMID: 31324866) involving in vitro phosphorylation of dephosphorylated protein mixtures with recombinant kinases followed by protein mass spectrometry-based quantification of phosphorylation events. Sugiyama et al did not do themselves a favor when they used the Motif-X algorithm to generate sequence preference motifs, but others have converted their data to proper position-specific matrices. Now it is unrealistic to expect the authors to recalculate predictions across the entire project, but this issue should be discussed. For the future, it would be of interest to compare the two approaches to profiling sequence preferences head-to-head perhaps in another paper.

Peptide libraries give even coverage across the kinome but, as the reviewer notes, can have modest signal-to-noise and report only what the active site sees; the Sugiyama et al. design presents protein substrates in native context and offers complementary evidence. Part of the head-to-head comparison the reviewer proposes has already been published, so we added a new Discussion subsection, “Motif models derived from peptide libraries and from in vitro phosphoproteomics”:

[...] An orthogonal strategy is to phosphorylate dephosphorylated proteome extracts with individual recombinant kinases and read out the products by mass spectrometry, as done by Sugiyama et al. for 385 kinases (1). This design presents kinases with protein substrates in native sequence context, and several groups have since converted the resulting data into position-specific matrices suitable for scoring. Poll et al. derived position-dependent Shannon information matrices for 384 kinases (2), and Invergo et al. built position-specific scoring matrices and naïve Bayes classifiers from the same source (3). [...] Poll et al. compared their mass-spectrometry-derived “logos” directly against the peptide-library motifs used here and reported a high degree of concordance, concluding that the best predictions are likely to come from combining both data types (2). A systematic head-to-head comparison of peptide-library and in vitro phosphoproteomic specificity models is an important future direction as an independent benchmarking enterprise.

2.

The use of discrete cut-offs or thresholds in the contextual filtering process can be regarded as a weakness. For example, it appears that for the Ochoa functional score a discrete cut-off of 0.5 was used. That means that a site with a score of 0.4999 is “out”, while a site with a score of 0.50001 is “in”. As understand it, a site with a score of 0.50001 gets the same emphasis as one with 0.999. Although the resource developed by the authors brings considerable value, future development of the tool should consider making the weightings probabilistic/Bayesian or at least transform the data to produce a continuously differentiable, monotonic weighting function. Although it is unrealistic to expect the authors to totally revise their calculations for the current publication, this issue should at least be discussed.

This is a fair critique; auditing it also surfaced the second implementation error, a data-derived cutoff of 0.375 that the pipeline had been enforcing instead of the stated 0.5, which we have corrected. We have added more detail to the Methods under “Phosphosite functional score”:

We downloaded the pre-computed human phosphosite functional scores and mapped them to our phosphosite identifiers (provided as Supp. Table 3). For each predicted KSI, we annotated the substrate site with its functional score and designated a site as “likely functional” if its score was ≥ 0.5 . This is the threshold recommended by the authors of the score, who report that at a 0.5 cut-off approximately 10% of the phosphoproteome is retained while 50% of true-positive functional sites are recovered, making it a useful cut-off for initial prioritization (4); comparable thresholds have been applied in subsequent work (5).

We also added a sensitivity analysis and a continuous confidence score. Methods, “Phosphosite functional score”:

Because any fixed threshold is arbitrary at its boundary, we additionally report a sensitivity analysis across the range 0.3-0.7 and release a continuous confidence score (see below).

The sensitivity analysis itself is reported in a new Results subsection, “Robustness of the filtering criteria”:

Sweeping the functional-score cut-off from 0.3 to 0.7 and the motif percentile from the 80th to the 99th (Supp. Figure 2) changes the size of the high-confidence set substantially [...] but leaves the qualitative conclusions intact. The proportion of retained predicted pairs involving a dark kinase varies only between 19.1% and 20.8% across that same functional-score range, and the number of kinases represented varies between 318 and 325.

The continuous score is described in a new Methods subsection, “Confidence scoring”:

Motif percentile and functional score are each passed through a monotonic, continuously differentiable logistic transform centred on the corresponding threshold, and localization evidence is expressed as a saturating function of the number of shared HPA compartments rather than a binary flag. [...] This is a transparent re-weighting rather than a calibrated probability: a fully probabilistic treatment would require labelled negative examples, which do not exist at kinome scale (3,6).

3.

One limitation of the approach described is that it excludes cell-type specific phosphorylation events because of the ‘consensus’ approach used. Cell-type specific function of specific cell types is largely the result of expression of a few specialized genes/proteins that account for a given cell’s unique behavior. Such proteins are often regulated by phosphorylation, which may not be predicted by the author’s model. A check of a few cell-type specific proteins indicates only previously curated KSIs, but no predicted KSIs. Of course the curated sites are already available through the resources mentioned by the authors. In contrast, the author’s model may be very useful in the setting of cancer, where the cells lose differentiated function. This issue might be mentioned in the Discussion.

This matches what we found once we checked it directly. We added a paragraph to the “Robustness of the filtering criteria” subsection:

Classifying substrates by HPA RNA tissue specificity, 70.4% of substrates with low tissue specificity gain at least one novel predicted interaction, falling monotonically to 57.8% for tissue-enhanced, 46.9% for group-enriched and 40.6% for tissue-enriched proteins.

We additionally report the likely causes with a new paragraph in the “Limitations” subsection:

[...] Three mechanisms contribute to this result. The specificity atlases score only phosphosites already reported in phosphoproteomics experiments, a literature dominated by immortalized and cancer cell lines, so a site never observed in those systems cannot enter the resource however good its motif match (7). Tissue-restricted proteins have sparser HPA annotation, so co-localization can fail for missing-data rather than biological reasons (8,9). [...] By the same logic, the resource is at its most useful in settings where differentiated programs are lost: cancer being the clearest case, and one where large phosphoproteomic cohorts already exist to test predictions against.

4.

It would be of interest to characterize the methodology with regard to relative information derived from the different data sources (e.g. in terms of ‘Shannon information content’). Note that it seems that the information sources may not be entirely orthogonal. For example, co-localization seems to be a factor in the functional score of Ochoa et al. and is of course central to HPA. So the simple question is whether HPA adds anything.

We computed the entropy of each evidence layer and the mutual information between them. On orthogonality, the 59 features of the Ochoa functional score span mass-spectrometry evidence, phosphosite regulation, structural environment and evolutionary conservation, and do not include subcellular localization (its “localization” features describe mass-spectrometric confidence in the site’s position within the peptide). We added a Methods subsection, “Information content of the evidence layers,” and the results to “Robustness of the filtering criteria”:

Across all motif-passing kinase-site pairs, subcellular co-localization carries the highest marginal entropy (0.896 bits, versus 0.418 bits for curated-site status and 0.430 bits for functional-score support). Pairwise mutual information between the three layers is very small [...], giving normalised redundancies of 0.19%, 0.16% and 1.95% respectively, and conditioning on the third indicator does not change this. The layers are therefore close to orthogonal, and each is doing distinct work: in a leave-one-out analysis, co-localization uniquely removes 47,029 kinase-site pairs that satisfy every other criterion, and omitting it would inflate the high-confidence set by 182%.

A new Discussion subsection, “Orthogonality and information content of the evidence layers,” interprets this:

This is expected from how the functional score is built. It is trained on 59 features spanning mass-spectrometry evidence, phosphosite regulation, structural environment and evolutionary conservation (4), and subcellular localization is not among them. The co-localization filter therefore contributes essentially independent evidence and does substantial work of its own, rather than rubber-stamping decisions already made by the other layers.

The analysis showed that HPA adds information that the other layers do not carry.

Reviewer 2

We are grateful for the reviewer’s careful, hands-on checking of our kinase-substrate predictions: it caught a significant error in the manuscript (specific comment 18). We address every point below in the order raised.

General comments

1.

One of the key goals of the KinaseCanvas website is to identify potential substrates and their phosphosites for novel (i.e., less characterized) kinases. This offers predictive capabilities generally

not available from other websites that track kinase-substrate connections. However, similar known and predictive information is available from www.phosphonet.ca, which is not cited.

We apologise for the omission and have added PhosphoNET to the Introduction:

PhosphoNET additionally provides sequence-based predictions of kinase-phosphosite pairings for 486 human protein kinases, using scoring matrices that span the -7 to +7 positions around the phosphoacceptor, alongside curated sites (10); the companion KiNECTOR resource (<http://www.kinector.ca>) aggregates over 25,000 human kinase-substrate-phosphosite relationships into navigable pathway maps.

We also compare our predictions with PhosphoNET's in our response to specific comment 17.

2.

One of the useful aspects of creation of the KinaseCanvas database is to identify potential sites of phosphorylation in a substrate for a kinase that demonstrated enhanced phosphorylation, but it is not known from the experiment where the substrate protein is phosphorylated. This might be evident from studies where non-specific phosphorylation is visualized with generic phosphotyrosine antibodies or band-shift of a substrate following incubation with a protein kinase with Phos-tagged SDS-PAGE gels and band shifts with a pan-specific antibody for the target.

This use case motivated the resource, and the portal did not serve it well enough. We have added this to the Results, "Interactive exploration and visualization":

Site-level interrogation of a specific pair: A pair mode restricts the view to a single kinase-substrate link and ranks the candidate sites on that substrate. This addresses the common experimental situation in which a kinase is known to phosphorylate a protein (e.g., from a band shift on a Phos-tag gel or a pan-phosphotyrosine blot) but the modified residue is unidentified.

3.

Overall, the KinaseCanvas website offers useful information to researchers in the field, although the underlying databases do not appear to be accessible in the publication and partially accessible in the KinaseCanvas website. The manuscript provides a good overview of the criteria that were generally applied in the selection of the kinase-substrate interactions that are posted on the KinaseCanvas website. However, the application of these criteria to specific kinase-substrate-phosphosite triads are not provided.

We have rewritten the Data Availability Statement to improve clarity:

Each released kinase-substrate-site triplet carries the individual evidence values behind its classification (motif percentile, the kinase's rank for that site, promiscuity index, phosphosite functional score, number of shared subcellular compartments and the composite confidence score), so the inclusion criteria can be checked case by case. [...] Supplementary Tables S1-S4 (S1 being the complete unfiltered prediction matrix) and the relaxed motif tier are deposited on Zenodo (<https://doi.org/10.5281/zenodo.22941318>) [...].

Specific comments

1.

Page 1, lines 42-44 -- "we distill millions of raw predictions into over 13,000 high-confidence novel interactions and integrate them with 8,365 curated predictions." Does the KinaseCanvas use just predicted data as opposed the empirical data? That is what the Abstract seems to imply as written, which should probably provide a bit more explanation of how the kinase-substrate interactions were filtered.

The wording was our error: the 8,365 (8,364 in the revised resource) were curated interactions, not predictions. We have rewritten the relevant sentence of the Abstract:

By applying stringent biological filters for subcellular co-localization and phosphosite functional importance, we distill tens of millions of raw predictions into 21,916 high-confidence novel interactions and integrate them with 8,364 curated interactions, yielding 30,280 distinct kinase-substrate pairs.

2.

Page 2, lines 11-13 -- “It has been estimated that roughly one-third of all eukaryotic proteins are phosphorylated in at least one cellular context, and that the human proteome may harbor on the order of 104 - 105 distinct phosphosites [1, 2].” Reference 1 is outdated, and Reference 2, which more recent, still underestimates the number of human phosphoproteins (~13,000) and phosphosites (~230,000 phosphosites). In the Vlastaridis (2027) publication, the authors correctly point out that high throughput mass spectrometry-based proteomics only tracks about 40-60% of the phosphosites. For example, out of over 4,000 HTP mass spectrometry studies documented in PhosphoSitePlus, about half of the 5000 best characterized functional phosphorylation sites are either not detected or only once or twice. Apparently, with a preponderance of basic residues flanking phosphosites (targeted for trypsin cleavage) and the high charge of phosphorylation sites (which is often clustered), many tryptic peptides have too high a charge to mass ratio for capture in the detection window of the mass spectrometer. Moreover, phosphosites are highly unstable due to phosphatase action during cell homogenization. Websites such as PhosphoSitePlus (www.phosphosite.org) and PhosphoNET (www.phosphonet.ca) list over 250,000 confirmed human phosphosites. Moreover, these phosphosites are distributed in more than 90% of the documented 21,000 human proteins. Thus, it would be much more accurate to state that over two-thirds of the human proteins are phosphorylated across more than a 250,000 phosphosites. Most of these sites will be rarely phosphorylated and at low stoichiometry, and likely inconsequential, but a small subset will be particularly influential in cellular regulation.

We removed reference 1 (Iakoucheva et al.). Rewritten opening of the Introduction:

Public repositories now catalogue well over 200,000 human phosphosites distributed across the large majority of the human proteome (11,12), and modelling that accounts for the incomplete and biased coverage of tryptic phosphoproteomics places the true totals higher still (12,13). Estimates of how many of these sites are individually functional are considerably lower (4,14,15), so the operative problem is not enumeration but prioritization.

3.

Page 2, lines 21-23 -- “Additional resources including Phospho.ELM, PhosphoNetworks and SIGNOR provide complementary KSI annotations derived from both focused and high throughput studies.” PhosphoELM has not been updated since September 2010. PhosphoNetworks is based on in vitro phosphorylation of recombinant proteins and does not identify the actual sites of phosphorylation. It also does not appear to be functionally operable at present for queries. I am surprised that www.phosphonet.ca is not listed as this website features both curated empirical and predicted Kinase-Substrate-Phosphosite relationships or www.kinector.ca, which graphically connects kinases and their substrates into maps that are much easier to navigate than shown in Signor-Phospho or for that matter KinaseCanvas.

These were addressed in the same Introduction paragraph, where the sentence on complementary resources now reads:

[...] although these differ substantially in scope and currency: Phospho.ELM has not been updated since 2010, and PhosphoNetworks reports kinase-substrate relationships measured on protein microarrays without resolving the modified residue (16,17).

4.

Page 2, lines 28-32 -- The authors describe the use of substrate peptide arrays and in vitro experiments to delineate substrate-specificity motifs for protein kinases, which can be used for prediction. This is a good starting place, but these motifs are typically derived from the mixing

of data from a large number of relatively poor substrates with a small number of high affinity substrates. Consequently, the identification of the critical flanking residues are rather diluted out, and there is a huge amount of overlap in consensus sequences. It is evident that more distant secondary binding sites are critical for kinase recognition as well as documented for MAP kinases and especially so with receptor-tyrosine kinases. Most kinase consensus motifs do not take into account flanking residues that are phosphorylated and critical for substrate recognition as known for casein kinases 1 and 2 and GSK3. This hierarchical phosphorylation is not really discussed in this manuscript.

The Limitations paragraph previously hinted at docking motifs and priming phosphorylation; it is now rewritten in full to address them directly. Discussion, Limitations:

Kinases are recruited to substrates through docking interactions distinct from the catalytic cleft, including the D- and DEF-motifs used by MAP kinases and the SH2- and PTB-domain interactions that bring receptor tyrosine kinase substrates to high effective local concentration (18). A peptide that scores poorly against a kinase's linear motif can therefore still be an efficient substrate once bound. Recognition is also frequently hierarchical: casein kinase 1 requires a phosphorylated residue at n-3, GSK3 at n+4, and casein kinase 2 can use either canonical or phospho-determinants, so a site's suitability depends on prior modification events that a static motif score does not represent (18).

5.

Page 3, lines 8-17 -- The authors point out the issue with reducing the number of predicted kinase-substrate interactions by applying filters such as co-location in the same tissue, cells and even compartments of cells. In addition to spatial relations, a kinase-substrate pair should also be temporally linked with parallel changes in kinase activity and substrate phosphorylation in response to diverse stimuli in vivo, shown by subsequent mass spectrometry or phosphosite-specific antibody analysis by Western blotting or antibody microarrays.

We agree that this would be an important next step for this resource. We have added a new limitation paragraph, in Discussion, Limitations:

Fourth, our filters are spatial but not temporal. Requiring a kinase and substrate to share a subcellular compartment is a necessary condition for a genuine interaction, not a sufficient one: a real kinase-substrate pair should additionally show coordinated behaviour over time, with substrate phosphorylation tracking kinase activity across stimuli, inhibitor treatments or genetic perturbation. Compendia of perturbation phosphoproteomics now make this tractable in principle (19,20), and adding a co-regulation layer [...] is the most obvious way to improve the resource.

6.

Page 3, Lines 47-49 -- The authors used the KiNET web portal with some 8,365 kinase-substrate interactions (1016 between kinases) as the starting place. This seems to represent a small subset of what appears to be over 25,000 kinase-substrate-phosphosite connection, most of which are documented in www.kinector.ca.

The 8,364 figure counts distinct kinase-substrate pairs. The same curated layer contains 16,360 distinct kinase-substrate-phosphosite relationships, which is the quantity comparable to the connection count the reviewer cites for KiNECTOR. Every entry is human, annotated to a reviewed (Swiss-Prot) UniProt accession, and deduplicated across PhosphoSitePlus, EPSD and iPTMnet as aggregated by KiNet; the remaining difference reflects the scope of that aggregation, and KiNECTOR does contain relationships our curated layer lacks (CDK1 at RANGAP1 S442, discussed under specific comment 17). KiNECTOR is now cited in the Introduction (general comment 1).

7.

Figure 1C -- The Alluvial diagram seems strange to me with most substrates identified as functional. The vast majority (>98 of the total number of known human phosphosites appear to

be non-functional. It would seem that established functionality is an major criteria for inclusion in the KinaseCanvas database.

We agree, the figure was misleading as drawn. It is not a sample of the human phosphoproteome; it is conditioned on sites that have already passed the motif filter and are present in curated phosphoproteomics databases. Rewritten Figure 1 caption:

Alluvial diagram tracking substrates through successive filtering stages: known vs. novel status, site-level support, and co-localization, to the final high-confidence set. The diagram is conditioned on substrates already retained by the initial motif and promiscuity filters and is not a sample of the human phosphoproteome.

The per-stage substrate counts, which the figure does not state, are now given in a new supplementary table, cited from Methods, “Integration of motif, localization and function”:

The effect of these filters on the substrate set is visualized as an alluvial diagram in Figure 1(C), with per-stage substrate counts given in Supp. Table 4.

Preparing that table also showed that the diagram’s last stage was drawn too loosely: it counted a substrate as retained if it had any supported site and, separately, any co-localized kinase, which kept 204 substrates whose supported sites and co-localized kinases never coincide. The diagram now applies the filter to the kinase-site pair as the pipeline does, so its final stage (2,471 substrates) equals the number the specificity atlas contributes to the resource (Figure 1B). The resource itself was not affected.

We also now cite the literature on weak functional constraint across phosphoproteomes in the Introduction (14), quoted under specific comment 2.

8.

Page 5, lines 16-18 -- It appears that some 15,644 distinct phosphosites involved in kinase-substrate relationships were identified. Were these all human phosphosites? Since some of the phosphosites were from other databases that were non-human originally, was the human equivalent phosphosite used? For example, a phosphosite may have been identified in mouse, but not previously reported in human cells. Was the human phosphosite sequence still used as a phosphosite?

This was an omission. New Methods subsection, “Species and isoform handling”:

The resource is restricted to human proteins (NCBI taxonomy 9606). Every protein in the curated KiNet table is a reviewed (Swiss-Prot) human UniProt entry, and we performed no cross-species orthologue mapping of our own, so a site enters the resource only if a contributing database reports it on the human protein. All phosphosite positions refer to the canonical UniProt sequence of the substrate: neither the curated table nor the specificity atlases use isoform-specific accessions. Every site is checked against that canonical sequence (see below), and the UniProt accession is released alongside every site so that positions can be verified independently.

9.

Page 5, lines 44-49 -- “Following the original studies, we focused on phosphosite-kinase pairs in which the site ranked within the top 10% of all phosphosites scored for that kinase (90th percentile) [10, 11]. This conservative cutoff retains sites that are strongly preferred by the kinase’s motif and reduces the inclusion of weak, low-specificity matches.” While this strategy has merit for focusing on the best fits between predicted Kinase-Substrate Interactions (KSIs), it is problematic when dealing with kinases that are more promiscuous than other kinases, and when secondary binding sites are also likely to be important. For example, many protein-tyrosine kinases are very promiscuous based on recognition motif, but SH2 and PTB domains may allow initial binding of what could be ranked as a poor substrate, but at infinite concentration (when bound to the kinase) becomes a highly phosphorylated substrate. Protein-serine/threonine kinases such as cAMP-dependent kinase, PKC- α , CaMK2, CK2A, ERK1/2, p38 MAPK have hundreds of known substrates. Ranking in the top 10% of scored phosphosites for a kinase may well exclude functionally important kinase-substrate interactions. It must be appreciated that during evolution

a given phosphosite may not necessarily be optimal for recognition by one kinase, but a set of different kinases from different signalling pathways.

The Limitations paragraph quoted under specific comment 4 now states the bias plainly:

Taken together, these mechanisms bias a top-percentile motif tier against precisely those kinases for which docking, rather than the catalytic cleft, dominates substrate selection – which is one reason we also release a continuous confidence score and a relaxed motif tier.

The relaxed tier is described in Methods, “Kinome-wide motif-based substrate predictions”:

Because a single stringent tier structurally excludes kinases whose specificity is dominated by docking interactions rather than by the catalytic cleft (see Discussion), we additionally release a relaxed tier comprising kinase-site pairs between the 75th and 90th motif percentiles that satisfy the same contextual criteria. This layer is released as a separate, labelled table (see Data Availability) and is not part of the high-confidence set.

10.

Page 6, lines 32-33 -- The authors acknowledge that the filtration of the KSI hits by co-localization had a limited effect. This is because the vast majority of proteins are expressed in the same cells as evidenced in the GEO database (and presented in www.transcriptonet.ca), and most proteins are found in the cytoplasm of cells, where they are at least initially made on ribosomes.

We quantified both sides of this in a new Results subsection, “Robustness of the filtering criteria”:

This filter is not trivially satisfied: the median gene has HPA annotations for two compartments and only 2.6% of annotated genes are assigned to five or more. [...] In a leave-one-out analysis, co-localization uniquely removes 47,029 kinase-site pairs that satisfy every other criterion, and omitting it would inflate the high-confidence set by 182%.

11.

Page 6, lines 50-52 -- “As expected, tyrosine sites were markedly enriched for high functional scores compared to serine or threonine sites (70.8% vs. 27.5% and 26.9% above the cutoff, respectively; Figure 2(B)).” Serine phosphosites are much less conserved in evolution than tyrosine and threonine phosphosites. However, there appears to be many more known threonine phosphosites (~20% of all sites) compared to tyrosine phosphorylation sites (~4% of all sites), and threonine phosphorylation sites are better conserved in evolution, especially when they are known to be functional phosphorylation sites. Dedicated protein-tyrosine kinases really emerged with multicellular eukaryotes, so evolutionary conservation of phosphothreonine sites is higher than tyrosine phosphosites. Most of the known tyrosine phosphorylation sites appear to be on receptor-tyrosine kinases and these arise from autophosphorylation, and are primarily for facilitating interaction with SH2- and PTB-domain containing adaptor or effector proteins. Consequentially, I am a little surprised that the Specificity Atlas came out this way. It seems to me that most protein-serine phosphorylation is devoted to facilitating the proteolytic degradation of proteins, so the exact placement of phosphoserine residues is less stringent. Since the phosphate group extends out further in phosphothreonine than phosphoserine, it has a better chance of further penetrating into a neighbouring protein region (internal to the protein, e.g., to disruption regulation, or externally for affecting protein-protein interactions) and causing disruption.

We answer this together with specific comment 16, which also concerns the tyrosine enrichment. With corrected thresholds, and now computed once per unique site rather than once per kinase-site pair, the figures are 36.1% tyrosine against 12.8% serine and 15.0% threonine, not 70.8%/27.5%/26.9%. Rewritten passage of Discussion, “Biological insights from the integrated network”:

We report this as a property of the assembled resource rather than as a claim about the phosphoproteome, because several factors plausibly contribute to it and we cannot separate them here. Much of the available phosphotyrosine data was generated using anti-phosphotyrosine immunoaffinity enrichment (21), a targeted design that both concentrates on sites of prior biological interest

and differs systematically from the untargeted enrichment used for serine and threonine sites. Phosphotyrosine is also the smallest and best-studied of the three classes, so its coverage is comparatively closer to saturation, and a substantial share of well-characterised tyrosine sites are receptor tyrosine kinase autophosphorylation sites whose documented function is to recruit SH2- and PTB-domain effectors (18). [...] the relative evolutionary conservation of phosphothreonine and phosphotyrosine sites remains an area of active disagreement (4,14). We therefore draw no evolutionary inference from this observation.

This difference is now backed by a statistical test the submitted version lacked, added to Methods, “Phosphosite functional score”:

The difference between residue classes is highly significant ($\chi^2 = 1848.1$, $df = 2$, $p < 2 \times 10^{-16}$; $n = 5,132$ tyrosine, 34,754 serine and 7,844 threonine sites).

On the relative conservation of phosphothreonine and phosphotyrosine sites, we are not in a position to adjudicate and, as quoted above, draw no evolutionary inference from our data.

12.

Page 8, lines 19-23 -The classification of a “dark kinase” is really related to time from its first discovery and its linkage to human disease. Moreover, gain of function is usually a criteria for consideration of a kinase drug target. Most kinases associated with disease are linked to cancer, and most oncogenes encode protein-tyrosine kinases as opposed to protein-serine/threonine kinases. There are only about 90 dedicated human protein-tyrosine kinases out of some 568 known and suspected human kinases, so it is not surprising that most “dark kinases” are serine/threonine-specific. The premise is that we can learn from the study of better known kinases to identify substrates for related, but less studied kinases. This is reasonable.

We agree, and appreciate that the reviewer finds the premise reasonable. Our Methods description of dark-kinase classification is already consistent with this framing (Methods, “Annotation of dark kinases”):

we used the DarkKinaseTools R package from the NIH Illuminating the Druggable Genome (IDG) programme to classify kinases as “understudied (dark)” or “well-studied (light)” based on publication counts, functional annotations and available reagents (22)

No change was needed: as the reviewer notes, with only ~90 dedicated tyrosine kinases, a serine/threonine skew among understudied kinases is expected.

13.

Page 9, lines 11-19 -- What is lacking is whether the phosphosite in the table is functional, especially when this is known in the literature, and what the functional effect of the phosphorylation actually is. The web portal also does not permit a query starting with a substrate to then identify upstream kinases, which is very unfortunate.

We answer comments 13-15 together, since all three concern the web portal.

14.

Page 11, lines 15-18 -- The authors have made much of the KSI database accessible to the broad community with the development and launch of KinaseCanvas Explorer. In looking at the KinaseCanvas website, four problematic features are: 1) how very promiscuous kinases are presented that have a large number of substrates. The maps are unwieldy and it is unclear how the order of their presentation is determined and shown. The icons are just circles, and do not offer information through shape and colour of different classes of proteins. 2) The underlying table provides phosphosite information, but it is ambiguous whether a phosphosite is known to be functional or suspected to be functional, and in what way it is functional (e.g., increased enzyme activity, altered subcellular location or protein interaction, increased degradation of protein). 3) Another problem is that the website is kinase-focused and not substrate-focused, which is entirely feasible with the underlying database that the authors have created. For example, these type

of issues are considered on the www.kinector.ca website, which also presents kinase-substrate connections, but not mentioned in this manuscript.

15.

Page 13, lines 18-20 -- “The Kinase Library web portal [10] serves as the primary interface for the Johnson et al. and Yaron-Barir et al. atlases, allowing users to score specific peptide sequences against kinase motifs.” It does not appear that KinaseCanvas offers any kind of actual quantitative data to evaluate how “good” a known or predicted substrate is.

The released interaction table did not contain the continuous scores for each K-S pair. We fixed that upstream, then rewrote the Results, “Interactive exploration and visualization,” in full. The relevant bullets now read:

Kinase- or substrate-first querying: The graph can be seeded from either a kinase or a substrate, with autocomplete on both fields. Seeding with a substrate returns all upstream kinases, curated and predicted, so the resource can be used to ask which kinases might act on a protein of interest as readily as the reverse. [...] Quantitative evidence on every interaction: The motif percentile, the kinase’s rank for that site among all kinases, the phosphosite functional score, the number of shared subcellular compartments and the composite confidence score are all displayed in a filterable table of the interactions in view and carried into every table export. A minimum-confidence slider filters predicted edges on the composite score. Explicit evidence provenance: Curated-site status and functional-score support are reported as separate fields, so a site known to be functional from the primary literature is distinguishable from one predicted to be functional, and the view can be restricted to curated pairs or to functional sites. Visual encoding: Node shape and colour encode role (kinase, substrate, or both), and for predicted edges the line thickness and opacity scale with the composite confidence score, so relative support is visible and not merely filterable.

On the question of in what way a site is functional: the phosphosite functional score is based on a gradient boosting model, trained on sites of known function, that aggregates 59 features grouped as mass-spectrometry evidence, phosphosite regulation, structural environment and evolutionary conservation into a single estimate of functional relevance (4). It indicates how likely a site is to be functional, not the mechanism; where a mechanism is known, it is described in the publications linked by PMID from each curated interaction’s detail panel.

On the promiscuous-kinase point specifically, the most connected kinase, CDK1, has 629 substrates in the revised resource. We added:

Legible neighbourhoods for promiscuous kinases: Highly connected kinases would otherwise render as unreadable hairballs; the most connected, CDK1, has 629 substrates. A seeded neighbourhood is therefore capped at 75 interactions by default, chosen under a stated ordering (curated first, then by descending composite confidence, ties broken alphabetically), with the count shown and the cap removable in one click. The cap does not affect the 63% of kinases, and 96% of all proteins in the resource, whose neighbourhoods contain 75 or fewer interactions.

Clicking any protein or interaction now also opens a detail panel. Results, “Interactive exploration and visualization”:

Click-to-inspect detail panel: Selecting a protein shows its role, UniProt link, numbers of substrates and upstream kinases, and its highest-ranked interactions. Selecting an interaction shows its evidence, sources and PMIDs, together with a per-site table giving each site’s flanking sequence, curation status, functional score, motif percentile, kinase rank and shared compartments, with curated sites listed first and the rest ranked by motif percentile.

We also added:

Site sequence context: Each site’s -7/+7 flanking sequence is shown with the phosphoacceptor highlighted, verified against the canonical UniProt sequence for every site in the resource.

These comments improved the resource more than any other part of the review.

16.

Page 14, lines 15-18 -- “The fact that 70.8% of tyrosine sites in our resource exceed the functional score threshold, compared with ~27% of serine and threonine sites, suggests that our pipeline preferentially retains tyrosine sites with substantial regulatory potential.” One of the reasons for why there seems to be a preponderance of functional tyrosine phosphosites is because many of the original mass spectrometry studies used preparations that were specifically enriched for tyrosine phosphorylated proteins through use of basic conditions with heat under which serine and threonine phosphorylation sites are less stable or used generic anti-phosphotyrosine antibodies. Thus, I suspect most of the human phosphotyrosine sites are now known, coupled with the keen interest in protein-tyrosine kinases because of their connection to oncogenesis following mutation. Many of the receptor-tyrosine kinases feature a large number of autophosphorylation sites that are functional in facilitating interactions with SH2 and PTB domain containing adaptor and effector proteins. Growth and differentiation factors often act through receptors that are protein-tyrosine kinases, so one can envision some evolutionary pressure for retention in diverse multicellular species. However, from my own analyses, phosphothreonine sites have better conservations in evolution.

Addressed together with comment 11 above.

17.

Page 14, lines 23-33 -- The authors noted that ABL1 is predicted to phosphorylate the ARID1A Y762 phosphosite, which has not been previously identified in curated databases such as PhosphoSitePlus. However, the www.phosphonet.ca website ranks the top 4 candidates out of 500 kinases in order of Arg, Brk, Abl and Tyro3. What makes the PhosphoNET predictions unique is that they are based on the primary amino acid sequence of the catalytic domains of 486 human protein kinases and scoring matrices that consider all of the amino acid residues from the -7 to +7 positions surrounding the phosphoacceptor site. Such kinase-substrate predictions have been available since 2017.

We thank the reviewer for checking this prediction against PhosphoNET. Its top-ranked candidates for ARID1A Y762 (Arg, Brk, Abl and Tyro3) include the two kinases our motif layer ranks highest (ABL2, i.e. Arg, and ABL1), which supports the motif-level result. However, ARID1A Y762 has a functional score of 0.38 and no curated support, so under the 0.5 threshold stated in the Methods it is not in the high-confidence set and is not an appropriate showcase. We replaced it with an example that is in the resource and whose prioritization rests on localization and functional score. Discussion, “Biological insights from the integrated network”:

The value of layering orthogonal evidence is illustrated by RANGAP1 S442, an experimentally observed mitotic phosphosite that lies in a cyclin-dependent kinase (Cdk) consensus motif but has no assigned upstream kinase in the curated databases we integrated (23). [...] Localization does separate them: CDK1 and CDK3 share the cytosol with RANGAP1, whereas CDK2 and GSK3B are annotated only to compartments that RANGAP1 does not occupy and MARK3 has no HPA annotation. The site also carries a high functional score (0.92).

As a check independent of our pipeline, KinaseNET and KiNECTOR both list RANGAP1 S442 as a known site of CDK1, one of the two kinases our pipeline retains (KiNECTOR also lists T409, the site curated in the databases we integrated); neither gives a ranking. OmniPath’s records for CDK1 at S442 (24,25) cite Swaminathan et al. (PMID 15037602), the study cited above, while its CDK2 and p38 (MAPK14) assignments for this site come from a computational prediction study (Linding et al., 2007; PMID 17570479). The experimental evidence for CDK1 at S442 therefore rests on a single study, and we present the site as a prioritised candidate rather than an established assignment. We could not repeat the reviewer’s PhosphoNET check for this site or for the example under specific comment 18: www.phosphonet.ca did not respond, or returned a “service unavailable” error, whenever we tried between 21 and 24 September 2026.

18.

Page 14, Lines 34-44 -- The authors suggest that KIFAP3 may be phosphorylated at S25 by NEK6. However, this is not a known phosphosite recorded in PhosphoSitePlus or Uniprot and not even predicted in PhosphoNET. In Uniprot, position 25 is a lysine residue, not a serine residue in the main isoform. In the Q92845-2 variant, there is a S25. This appears as S69 in the longer KIFAP3 isoform, and is a predicted but not confirmed phosphosite in PhosphoNET, and not identified in PhosphoSitePlus. In PhosphoNET NEK6 is not in the list of top 50 predicted kinases for the S69 site, although NEK10 is ranked the second best after CHK2. NEK6 has relatively few known substrates as documented in www.kinaseset.ca and www.kinector.ca. The presence of a glutamic acid at the -3 residue in the S69 site would appear to serve as a negative determinant for NEK6 recognition, whereas an arginine residue at the -3 residue is a strong positive determinant. On the PhosphoSitePlus website the consensus logo based on 52 aligned substrate phosphosite sequences supports a leucine residue at the -3 position as a strong positive determinant, and a glutamic acid residue at the -3 position also appears as a negative determinant. Consequently, the KIFAP3 S69/S25 site really does not look attractive as a substrate for NEK6. Ideally, if such a prediction is offered in this manuscript, there should be an effort to validate this experimentally or show some independent support for this from the literature.

The reviewer is right that this prediction should not have been made, and we are grateful for the care taken in checking it by hand. The cause was worse than the isoform-numbering issue the reviewer reasonably inferred, and we describe it in full because it affected the resource beyond this one example.

The site does not exist on KIFAP3 at all. Our identifier-harmonisation code resolved gene symbols through an alias lookup and kept every match when an alias mapped to more than one gene. “SMAP” is an alias of C11orf58 (UniProt O00193), of KIFAP3 (via “Smg GDS-associated protein”) and of BRD8, so a single phosphosite on O00193, S25, was emitted under three different substrate names while keeping O00193’s residue numbering. The specificity atlas contains no S25 entry for KIFAP3; only S60 and S102. The reviewer’s observation that UniProt Q92845 position 25 is a lysine is correct, and was the first sign of this.

We have added a new Methods subsection, “Substrate and kinase identity resolution,” describing the fix:

We therefore resolve substrate identity from the UniProt accession supplied by the specificity atlases, falling back to the gene symbol only where it maps unambiguously to a single gene. Kinase identity is resolved from the atlas column names [...] against a curated kinome reference table, then against unambiguous aliases, with a small explicit override list for names that neither source covers; the pipeline halts if any atlas kinase name remains unresolved rather than silently discarding it. Every released kinase-substrate-site triplet is checked automatically against the source atlas to confirm that the site belongs to the accession it is reported under. Every site is additionally checked against its canonical UniProt sequence, which also yields the -7/+7 flanking sequence shown for each site (verified for 95.4% of sites; unverified sites are left without one rather than shown an unverified guess).

The mistake was general, so we audited the whole resource. It affected 12 of 10,911 substrate accessions. Of 396 atlas kinase names, 51 matched neither an official symbol nor an alias and were dropped without warning, and 13 were ambiguous, fanning out onto 15 additional genes. “ABL” resolved to ABL1, MTTP and DSP-AS1, so a tyrosine-kinase specificity profile was attributed to a lipid-transfer protein and an antisense RNA. With these corrections, integration raises the dark-kinase share of pairs from 3.4% to 15.7% (11.2% in the submitted version), and every figure and statistic that depends on it has been updated.

The case study has been replaced. We rewrote the relevant paragraph of Discussion, “Biological insights from the integrated network”:

Similarly, our resource illuminates potential regulatory roles for dark kinases. MAPK15 (ERK8) is classified as understudied by the IDG programme (22) and has few annotated substrates despite belonging to a well-characterised, proline-directed kinase family. Our pipeline nominates RING1, the E3 ubiquitin ligase subunit of the Polycomb repressive complex 1, as a candidate substrate at S38. The prediction satisfies every criterion in the pipeline and is internally coherent with what is known about the kinase: MAPK15 is the top-ranked kinase of the entire kinome for this

site (rank 1, 99.0th percentile), the site carries a high functional score (0.89), MAPK15 and RING1 share two annotated compartments (cytosol and nucleoplasm), and the local sequence (DGTEIAV-S-PRSLHSE) is a canonical serine-proline motif of the kind MAPK-family kinases require. Six of the seven kinases that score at or above the 90th percentile for this site are retained (PINK1, ranked third, shares no compartment with RING1), and MAPK15 ranks first among them, well ahead of ERK1 (MAPK3; rank 2, 93.5th percentile). S38 is an experimentally observed phosphosite that currently has no assigned upstream kinase in the curated databases we integrated, which is precisely the gap this resource is intended to fill. We present it as a prioritised hypothesis for experimental testing rather than as an established mechanism.

MAPK15 -> RING1 S38 is not listed in KinaseNET or KiNECTOR, as expected for an interaction that is novel with respect to curated resources, and we found no independent experimental support for it in the literature, HIPPIE or OmniPath. We therefore present it strictly as a prioritised hypothesis, and rest the argument for the pipeline on the RANGAP1 example (specific comment 17), where the prioritized kinase is independently supported.

This flanking-sequence context, worked out by hand for the case study above, is now surfaced for every site in the Explorer itself, verified against the canonical UniProt sequence. See the Results passage quoted under specific comment 15.

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